

HealthStat: Healthcare Analytics Dashboard in Power BI

Project Overview

HealthStat is an interactive Power BI dashboard designed to analyze healthcare performance metrics for elective hip replacement inpatient stays in New York State.

The dashboard enables users to evaluate hospital performance, compare costs, examine length of stay (LOS) trends, and identify healthcare utilization patterns across hospitals.

Business Problem

Healthcare administrators need visibility into operational efficiency and cost performance across hospitals. This project provides a centralized dashboard to:

- Compare hospital length of stay metrics
- Analyze discharge volumes
- Evaluate cost per discharge
- Identify high-performing and low-performing hospitals
- Support data-driven healthcare decision making

Dataset

The project utilizes healthcare inpatient stay data related to elective hip replacement surgeries in New York State.

Key metrics analyzed include:

- Average Length of Stay (LOS)
- Cost per Discharge
- Total Discharges
- Hospital-Level Performance Indicators

Dashboard Pages

1. Home Page

Landing page providing navigation to all dashboard sections.

2. LOS Comparison

Analyzes:

- Average Length of Stay by hospital
- Highest LOS hospitals
- Lowest LOS hospitals
- Trends and variation across providers

3. Cost Comparison

Examines:

- Average Cost per Discharge
- Cost vs LOS relationships
- High-cost and low-cost facilities
- Hospital benchmarking

4. Hospital Profile

Provides a detailed hospital-level breakdown including:

- Average LOS
- Average Cost per Discharge
- Total Discharges
- Patient disposition analysis
- Monthly discharge trends
- Diagnosis-related discharge distributions

Key Features

- Interactive filtering and slicing
- KPI cards
- Scatter plot analysis
- Trend analysis
- Hospital benchmarking
- Drill-down capability
- Comparative performance analysis

Tools Used

- Power BI Desktop
- Power Query
- DAX
- Data Modeling
- Data Visualization

Example KPIs

The dashboard tracks:

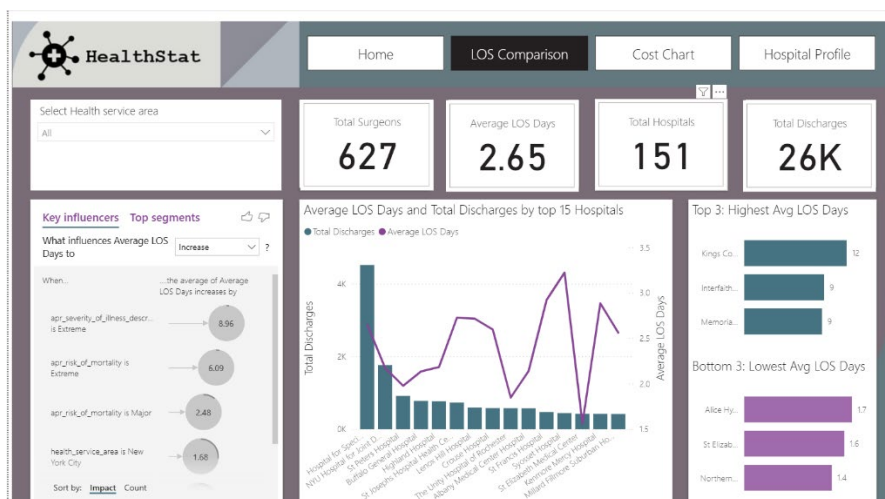
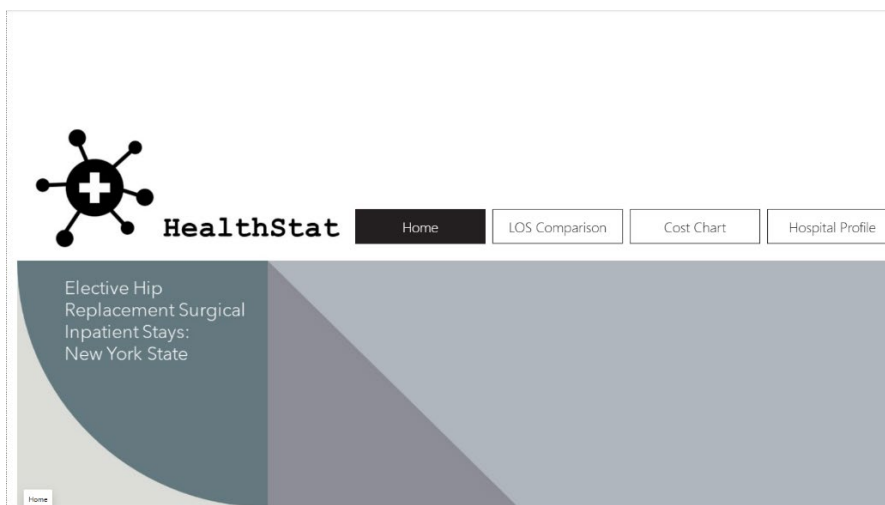
KPI	Description
Average LOS	Average inpatient stay duration
Cost per Discharge	Average cost associated with each discharge
Total Hospitals	Number of hospitals analyzed
Total Discharges	Total patient discharges

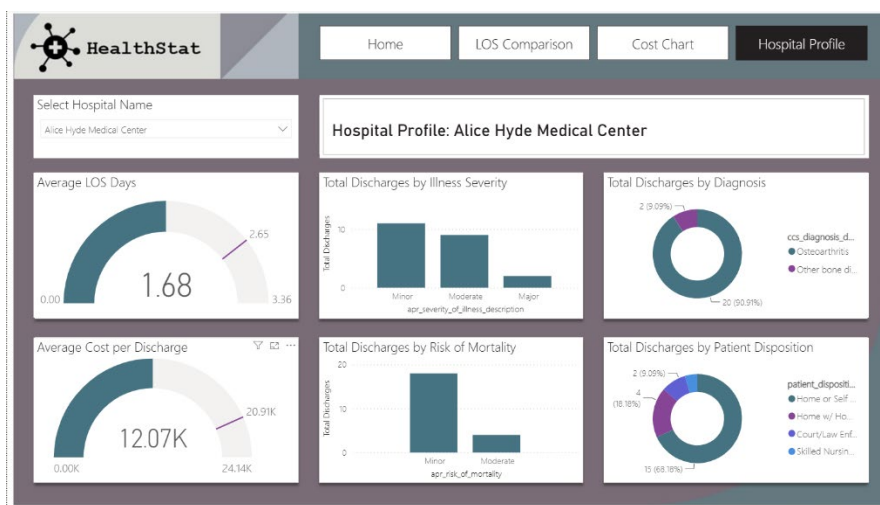
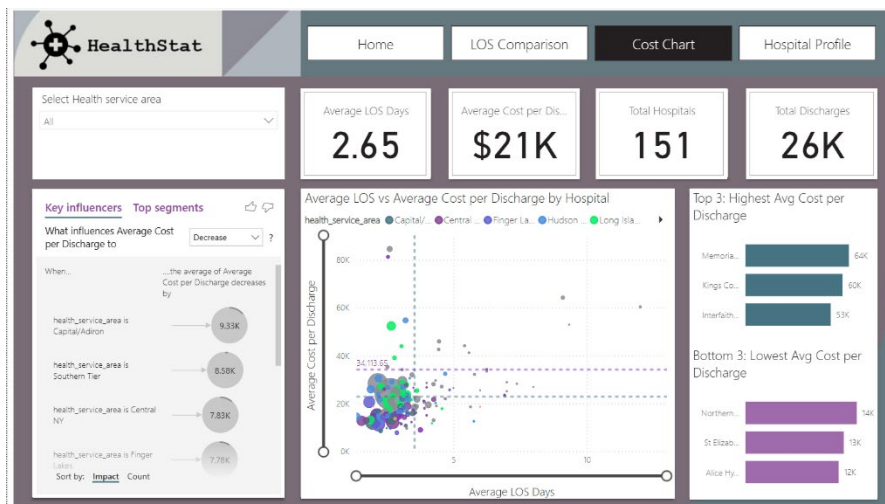
Insights Generated

The dashboard helps identify:

- Hospitals with unusually high LOS
- Cost outliers
- Relationships between LOS and treatment costs
- Discharge volume patterns
- Facility-level performance differences

Screenshots





Skills Demonstrated

- Data Cleaning
- Data Transformation
- Data Modeling
- DAX Calculations
- KPI Development
- Healthcare Analytics
- Dashboard Design
- Business Intelligence Reporting